

Windowed Phase Conditioning and Lag-One Covariance of Consecutive Prime Gaps

Reddouane Berramdane

Independent researcher

7 October 2026

v1.0 tag `paper2-v1.0`; source/data commit `bb3548c`

Preprint DOI: [10.5281/zenodo.23222536](https://doi.org/10.5281/zenodo.23222536)

Abstract

We study the lag-one covariance of consecutive prime gaps $g_n = p_{n+1} - p_n$ in the window $[10^{10}, 3 \times 10^{10})$, after local detrending and conditioning on the opening-prime residue modulo a primorial M . For each of seven moduli $M \in \{6, 30, 210, 2310, 30030, 510510, 9699690\}$ we obtain an exact decomposition $A + X + Y + D = \text{Cov}$ that closes to within 10^{-6} . The residual correlation ρ_{resid} remains negative at all seven scales, decreasing in magnitude from -0.0247 to -0.0103 . The between-phase variance grows monotonically with M , accounting for up to 3.74% of $\text{Var}(e)$. The non- D terms account for 60.4% of the lag-one covariance at the largest tested modulus, while between-phase structure accounts for 3.74% of the one-point variance. We report block-bootstrap 95% intervals for ρ_{resid} that exclude zero at every scale. The results are finite-range and do not claim asymptotic laws.

1 Introduction

Let p_n be the n -th prime and let

$$g_n = p_{n+1} - p_n$$

be the n -th consecutive prime gap. The distribution and size of prime gaps have been studied extensively since Cramér (1936), with major contributions by Gallagher (1976) and Granville (1995), and more recent work on arithmetic structure and correlations by Holt (arXiv:2405.03540), the Lemke Oliver–Soundararajan work on consecutive-prime biases (LOS, arXiv:1603.03720), the Montgomery–Soundararajan framework (MS, arXiv:math/0409258), Ash–Beltis–Gross–Sinnott (2011), and Banks–Ford–Tao (arXiv:1908.08613). Much less is known, however, about the serial dependence of consecutive prime gaps after local scale effects have been removed.

In Berramdane (2026a, DOI 10.5281/zenodo.23162269), the first paper in this series, we studied locally detrended prime-gap windows and reported persistent negative lag-one dependence up to $N = 10^9$. The present paper extends that investigation to the range

$$[10^{10}, 3 \times 10^{10})$$

and introduces a sieve-phase conditioning step. The central observable is the lag-one covariance of consecutive prime gaps after local detrending and after conditioning on the opening-prime residue modulo a primorial M .

The data set analyzed here contains

$$n = 844,953,415$$

gaps in the interval $[10^{10}, 3 \times 10^{10})$. For the detrended gap sequence we find

$$\text{Var}(e) = 438.670, \quad \text{sd}(e) = 20.944,$$

with

$$d = L - \text{sd} = 2.631, \quad |\text{Cov}|/(L/2) = 0.933,$$

and

$$\rho_1 = -0.025078.$$

These values reproduce the frozen predictions made before the 10^{10} run.

The main technical device is an exact covariance decomposition. For each primorial modulus

$$M \in \{6, 30, 210, 2310, 30030, 510510, 9699690\},$$

we decompose the lag-one covariance into four terms

$$A + X + Y + D = \text{Cov},$$

where A is the phase component, X and Y are cross terms, and D is the residual term. The decomposition closes exactly to within 10^{-6} for every M in the table. The residual correlation ρ_{resid} remains negative for all seven moduli, ranging from -0.024703 at $M = 6$ to -0.010306 at $M = 9699690$. Thus phase conditioning reduces the residual term D from -10.83 at $M = 6$ to -4.35 at $M = 9699690$ and redistributes the covariance among A , X , Y , D , while the total remains fixed at -11.001171 . The residual correlation remains negative at all tested scales. At the largest tested modulus, the non- D terms $A + X + Y$ account for 60.4% of the lag-one covariance $A + X + Y + D$, while phase structure accounts for 3.74% of the one-point variance.

This paper makes five contributions. First, it reproduces the frozen 10^{10} predictions and confirms the stability of the negative lag-one dependence at the larger scale. Second, it gives an exact sieve-phase decomposition of the lag-one covariance. Third, it reports the decomposition across seven primorial scales. Fourth, it studies the variance structure of the gap sequence through between-phase and within-phase components. Fifth, it states the limitations of the present analysis and distinguishes the observable studied here from related recurrence-statistics approaches.

We do not claim that phase conditioning explains all dependence, nor that the residual converges to zero, nor that the present results prove the Hardy–Littlewood or Montgomery–Soundararajan laws. The aim is narrower: to measure, decompose, and stress-test the lag-one covariance of consecutive prime gaps in a controlled windowed setting, and to report what remains after the dominant sieve-phase component has been separated.

The paper is organized as follows. Section 2 defines the data and windowing procedure. Section 3 presents the sieve-phase decomposition. Section 4 gives the covariance decomposition. Section 5 reports scale validation. Section 6 analyzes the variance structure. Section 7 discusses theory and records a finite inclusion–exclusion diagnostic as future work. Section 8 states limitations. Section 9 records the distinction from Murray (SSRN 6947578, SSRN 7426882).

2 Data and Windowing

2.1 Data set

We analyze consecutive prime gaps in the interval

$$[10^{10}, 3 \times 10^{10}).$$

Let p_n denote the n -th prime. The data set contains

$$n = 844,953,415$$

consecutive prime gaps

$$g_n = p_{n+1} - p_n.$$

The center of the interval in log scale is

$$\log p_{\text{center}} = 23.575157.$$

The detrended gap variance is

$$\text{Var}(e) = 438.670412,$$

with standard deviation

$$\text{sd}(e) = 20.944.$$

The lag-one covariance of the detrended gap is

$$\text{Cov}(e_n, e_{n+1}) = -11.001171,$$

and the lag-one correlation is

$$\rho_1 = -0.025078.$$

These values reproduce the frozen predictions made before the 10^{10} run.

2.2 Local detrending

We follow the windowed local detrending procedure frozen in the analysis protocol and used in Berramdane (2026a). The detrending removes the local scale trend within each window so that the remaining signal reflects serial dependence rather than slow variation in gap size. No parameter of the detrending is changed in the present study. The residual gaps are the input to the covariance and correlation analyses reported below.

The variance and covariance values reported below are those of the detrended residual e_n , not of the raw gap g_n . The relation is

$$\text{Var}(e) = \text{Var}(g) - \beta^2 \text{Var}(\log p).$$

For the 10^{10} window we obtain

$$\text{Var}(g) = 438.765647, \quad \text{Var}(\log p) = 0.095254,$$

$$\beta = 0.999903, \quad \beta^2 \text{Var}(\log p) = 0.095235,$$

so $\text{Var}(g) - \beta^2 \text{Var}(\log p) = 438.670412$, closing with the measured $\text{Var}(e) = 438.670412$ to within 4×10^{-13} .

2.3 Phase conditioning

For each primordial modulus

$$M \in \{6, 30, 210, 2310, 30030, 510510, 9699690\},$$

we condition on the opening-prime residue

$$r = p_n \bmod M.$$

This partitions the gaps into $\varphi(M)$ phase classes. Table 1 lists, for each M , the average number of gaps per class n_c and the minimum class count min_class .

The smallest class count occurs at $M = 9699690$, where $\text{min_class} = 433$. All classes are non-empty and sufficiently populated for the covariance decomposition and the bootstrap analyses reported in the following sections.

Table 1: Phase class sizes.

M	n_c	$\min_class$
6	422,476,707.50	422,475,051
30	105,619,176.88	105,617,814
210	17,603,196.15	17,601,182
2310	1,760,319.61	1,758,565
30030	146,693.30	146,003
510510	9,168.33	8,936
9699690	509.35	433

2.4 Windowing and resampling

The 10^{10} data set is divided into 40 contiguous blocks for the block bootstrap. Confidence intervals for ρ_{resid} are obtained by resampling these blocks. The bootstrap uses independent seeds for each M and is described in the frozen protocol. The resulting 95% intervals are reported in Section 4.5.

2.5 Reproducibility

All results in this paper are computed from the frozen 10^{10} data set. No value is reported without a corresponding run. The file `phase_decomposition_v2.csv` contains the full table of M , the covariance decomposition, the variance split, and the bootstrap intervals.

3 Sieve-phase decomposition

3.1 Setup

Let p_n be the n -th prime and let $g_n = p_{n+1} - p_n$. For a primorial modulus

$$M \in \{6, 30, 210, 2310, 30030, 510510, 9699690\},$$

define the phase class of the n -th gap by

$$r_n = p_n \bmod M.$$

Each M is squarefree and composed of the first k primes, so the set of admissible residues r_n is exactly $(\mathbb{Z}/M\mathbb{Z})^\times$, of size $\varphi(M)$.

We work with the locally detrended gap

$$e_n = g_n - (\alpha + \beta \log p_n),$$

where α and β are the ordinary-least-squares coefficients of g_n on $\log p_n$. On a single window, this OLS detrending coincides with the "windowed local detrending" of Section 2.2. For the 10^{10} window we obtain

$$\alpha = 0.002492, \quad \beta = 0.999903.$$

3.2 Phase-conditional means

For each admissible residue r define the phase-conditional mean

$$\mu(r) = \mathbb{E}[e_n \mid r_n = r],$$

estimated by the sample mean of e_n over the gaps with $r_n = r$. The phase-conditional mean captures the systematic offset of the detrended gap as a function of the opening-prime residue.

3.3 Between-phase and within-phase variance

By the law of total variance,

$$\text{Var}(e) = \text{Var}(\mu(r_n)) + \mathbb{E}[\text{Var}(e_n \mid r_n)].$$

The first term is the between-phase variance; the second is the within-phase variance. In the CSV these are `var_between` and `var_within`.

For the 10^{10} window, $\text{Var}(e) = 438.670412$ and the split is reported in Table 2:

Table 2: Variance split by modulus.

M	$\varphi(M)$	<code>var_between</code>	<code>var_within</code>	<code>var_between</code> / <code>var_e</code>
6	2	0.157635	438.512777	0.00036
30	8	1.283949	437.386463	0.00293
210	48	3.879416	434.790995	0.00884
2310	480	6.750875	431.919537	0.01539
30030	5760	9.793090	428.877321	0.02232
510510	92160	12.722069	425.948343	0.02900
9699690	1658880	16.402117	422.268294	0.03739

Two observations:

1. The split closes exactly: `var_between` + `var_within` = 438.670412 for every M .
2. `var_between` grows monotonically with M , from 0.158 at $M = 6$ to 16.402 at $M = 9699690$. At the largest modulus, phase conditioning accounts for approximately 3.74% of the total variance of the detrended gap.

The within-phase variance declines correspondingly. This is a purely arithmetic effect: refining the modulus resolves more of the residue-class structure of the opening prime, and part of the gap-size variance is explained by that structure.

3.4 Class population

All classes are non-empty. The minimum class count across the seven moduli is reported in Table 1 (Section 2.3). The sparsest case is $M = 9699690$, where the smallest class contains 433 gaps. This is sufficient for the covariance decomposition and for the block bootstrap reported in Section 4.5.

3.5 What this section does not claim

The decomposition is an algebraic identity. It partitions the variance into a between-phase and a within-phase term; it does not attribute a causal mechanism to either. The phase-conditional mean $\mu(r)$ is a computed quantity, not a modelled one, and no asymptotic claim is made about its behaviour as $M \rightarrow \infty$.

4 Covariance decomposition

4.1 Decomposition identity

Write the detrended gap as

$$e_n = \mu(r_n) + u_n, \quad \mathbb{E}(u_n \mid r_n) = 0,$$

where u_n is the within-phase residual. The lag-one covariance decomposes as

$$\text{Cov}(e_n, e_{n+1}) = A + X + Y + D,$$

with

$$A = \mathbb{E}[\mu(r_n) \mu(r_{n+1})],$$

$$X = \mathbb{E}[\mu(r_n) u_{n+1}],$$

$$Y = \mathbb{E}[u_n \mu(r_{n+1})],$$

$$D = \mathbb{E}[u_n u_{n+1}].$$

All four terms are computed from the same data; the identity is exact by construction, not an approximation.

4.2 Values

The observed components, at 10^{10} and 6 decimal places:

We also report the non- D share of the lag-one covariance,

$$S(M) = \frac{A(M) + X(M) + Y(M)}{\text{Cov}(e_n, e_{n+1})} = 1 - \frac{D(M)}{\text{Cov}(e_n, e_{n+1})}. \quad (1)$$

Because Cov and D are both negative, $S(M)$ is positive and increases with M . It is an algebraic decomposition share, not a causal fraction of dependence removed by conditioning.

Table 3: Covariance decomposition components. The sum $A + X + Y + D$ equals Cov exactly for every M and is not repeated as a column. The last column reports $S(M) = (A + X + Y)/\text{Cov}$, the non- D share.

M	A	X	Y	$X + Y$	D	Cov	$S(M)$
6	-0.014798	-0.011110	-0.142602	-0.153712	-10.832660	-11.001171	1.5%
30	-0.086958	-0.118493	-1.170132	-1.288625	-9.625587	-11.001171	12.5%
210	+0.138938	-0.055117	-3.084120	-3.139237	-8.000871	-11.001171	27.3%
2310	+0.670972	-0.035007	-4.876237	-4.911244	-6.760899	-11.001171	38.5%
30030	+1.369692	-0.029422	-6.575378	-6.604800	-5.766063	-11.001171	47.6%
510510	+2.127980	-0.028428	-8.085351	-8.113779	-5.015372	-11.001171	54.4%
9699690	+2.898121	-0.031791	-9.515775	-9.547566	-4.351726	-11.001171	60.4%

4.3 Qualitative features

Three features stand out.

(i) Sign change of A . The phase component A is small and negative for $M = 6$ and $M = 30$, then changes sign between $M = 30$ and $M = 210$, and grows monotonically for $M \geq 210$, reaching 2.898 at $M = 9699690$.

(ii) Dominance of D . The residual term D dominates the decomposition at small M ($D = -10.833$ out of $\text{Cov} = -11.001$ at $M = 6$), and decreases in magnitude as M grows, reaching -4.352 at $M = 9699690$. The residual remains negative at every tested modulus.

(iii) Co-dominance of Y and D at large M . At small M ($M = 6, 30$) the residual term D dominates the decomposition. At larger M the cross term Y grows in magnitude and becomes comparable to D : at $M = 9699690$, $Y = -9.516$ while $D = -4.352$. The two co-dominate at the largest modulus tested. The term X remains small at every M . The phase component A changes sign between $M = 30$ and $M = 210$ and grows monotonically for $M \geq 210$.

4.4 Residual correlation

Define the residual lag-one correlation

$$\rho_{\text{resid}}(M) = \frac{D(M)}{\text{var_within}(M)}.$$

The observed values are reported in Table 4:

Table 4: Residual lag-one correlation.

M	ρ_{resid}
6	−0.024703
30	−0.022007
210	−0.018402
2310	−0.015653
30030	−0.013445
510510	−0.011775
9699690	−0.010306

The residual correlation remains negative at all seven tested scales. Its magnitude declines from 0.0247 at $M = 6$ to 0.0103 at $M = 9699690$ — a reduction of roughly 58%. Phase conditioning therefore reduces the residual term D from -10.83 at $M = 6$ to -4.35 at $M = 9699690$, while the total covariance $A + X + Y + D$ remains fixed at -11.001171 . The redistribution among the four components does not eliminate the residual dependence within the window.

The CSV also contains a small-sample-corrected variant $\rho_{\text{resid}}^{\text{corr}}(M)$, which differs from the raw value only at the largest modulus: -0.010326 versus -0.010306 at $M = 9699690$. The correction is negligible at all tested scales and the paper reports the raw values.

4.5 Block bootstrap

Confidence intervals for ρ_{resid} are obtained by a moving-block bootstrap with 40 contiguous blocks and 1000 replicates, with an independent seed for each M . The 95% intervals are reported in Table 5:

Table 5: Block-bootstrap 95% confidence intervals for ρ_{resid} .

M	ρ_{resid}	95% CI	width
6	−0.024703	[−0.024868, −0.024544]	0.000324
30	−0.022007	[−0.022138, −0.021864]	0.000274
210	−0.018402	[−0.018511, −0.018283]	0.000228
2310	−0.015653	[−0.015750, −0.015551]	0.000199
30030	−0.013445	[−0.013535, −0.013352]	0.000183
510510	−0.011775	[−0.011856, −0.011691]	0.000165
9699690	−0.010306	[−0.010379, −0.010226]	0.000153

All seven marginal 95% bootstrap intervals exclude zero. The point estimates also decrease monotonically across the tested modulus ladder. We do not claim a familywise test of monotonicity.

4.6 What this section does not claim

A , X , Y , D are conditional components computed from the same data. The decomposition does not establish a causal role for the phase component. It establishes that, under the present

conditioning, the residual term D remains negative at all seven tested moduli, and that the total covariance is redistributed across A , X , Y , D as M grows: A becomes increasingly positive, Y becomes increasingly negative, and D decreases in magnitude. The sum $A + X + Y + D$ is unchanged.

No claim is made that $\rho_{\text{resid}}(M) \rightarrow 0$ as $M \rightarrow \infty$. The seven moduli are a finite ladder; extrapolation beyond $M = 9699690$ is not supported by the present data.

A note on the mechanical origin of Y . The next phase class satisfies $r_{n+1} = r_n + g_n \bmod M$, so the gap g_n determines r_{n+1} exactly. Part of the transfer of covariance from D to Y as M grows is therefore a re-distribution within the accounting identity, not a physical removal of dependence. The total $A + X + Y + D$ is fixed at -11.001171 for every M ; what changes is its distribution across the four components. This identity is also the reason that Y is conceptually related to the consecutive-residue transition viewpoint discussed in Section 7.

5 Scale validation

We repeated the frozen analysis on the disjoint opening-prime window

$$[10^{10}, 3 \times 10^{10}),$$

containing 844,953,415 gaps.

Before the run, the following expected ranges were specified in the frozen protocol:

- $\text{Var}(e) \in [438, 445]$,
- $d = L - \text{sd}(e) \in [2.5, 2.7]$,
- $\text{Cov}(e_n, e_{n+1}) \in [-11.8, -11.0]$,
- $\rho_1 \in [-0.0269, -0.0248]$.
- $\text{sd}(e) \in [20.9, 21.1]$,
- $|\text{Cov}|/(L/2) \in [0.93, 1.00]$.

The observed values were

$$\text{Var}(e) = 438.670412, \quad \text{sd}(e) = 20.944,$$

$$d = L - \text{sd}(e) = 2.631, \quad \text{Cov}(e_n, e_{n+1}) = -11.001171,$$

$$\rho_1 = -0.025078, \quad |\text{Cov}|/(L/2) = 0.933285.$$

All six values fall inside their pre-specified ranges. Two of them lie close to a range boundary: $\text{Cov} = -11.001171$ lies 0.001171 above the lower edge of $[-11.8, -11.0]$, and $|\text{Cov}|/(L/2) = 0.933285$ lies 0.003285 above the lower edge of $[0.93, 1.00]$. This agreement is a finite-range out-of-sample consistency check, not an asymptotic validation. The pre-registered ranges were specified for the detrended quantities in the frozen protocol file `extensions/ie_test/future_work/protocol.md`, committed at source/data commit `bb3548c` before the 10^{10} run.

The window is defined by the opening prime p_n ; the closing prime of the final gap may exceed the upper endpoint. The two-pass streaming implementation used 64-bit prime coordinates and did not reconstruct primes from floating-point logarithms. This is the correction that resolved the earlier overflow at $p > 2^{32}$.

6 Variance structure

6.1 Setup

Let e_n denote the detrended gap and write the conditional decomposition

$$e_n = \mu_n + u_n, \quad \mathbb{E}(u_n \mid p_n \bmod M) = 0,$$

where $\mu_n = \mathbb{E}(e_n \mid p_n \bmod M)$ is the phase-conditional mean and u_n is the within-phase residual.

By the law of total variance,

$$\text{Var}(e) = \text{Var}(\mu_n) + \mathbb{E}[\text{Var}(u_n \mid r)].$$

The variance split `var_between` + `var_within` = `var_e` is reported in Table 2 (Section 3.3). This section interprets that split and records its relation to the covariance decomposition of Section 4.

6.2 Monotone growth of the between-phase variance

The between-phase variance `var_between` grows monotonically with M across the tested ladder, from 0.158 at $M = 6$ to 16.402 at $M = 9699690$. At the largest modulus this accounts for approximately 3.74% of the total detrended variance $\text{Var}(e) = 438.670412$. The within-phase variance declines correspondingly.

The growth is a purely arithmetic effect of the conditioning. As M increases, more of the residue-class structure of the opening prime is resolved, and a larger fraction of the gap-size variance is attributed to phase. No claim is made that this fraction converges to a limit as $M \rightarrow \infty$; the seven moduli form a finite ladder.

6.3 Relation to the covariance decomposition

Section 4 decomposes the lag-one covariance as

$$\text{Cov}(e_n, e_{n+1}) = A + X + Y + D,$$

where A is the phase component, X and Y are cross terms, and D is the residual term.

Both decompositions arise from the same phase conditioning. The between-phase variance `var_between` and the phase covariance component A capture the contribution of phase structure to two different second-order statistics:

- `var_between` is a one-point statistic: it measures the dispersion of the phase-conditional means $\mu(r)$ around the global mean.
- $A = \mathbb{E}[\mu(r_n)\mu(r_{n+1})]$ is a two-point statistic: it measures the lag-one autocovariance of the phase-conditional means.

The two quantities are not equivalent. A phase structure with large one-point dispersion need not produce a large lag-one autocovariance, and vice versa. The observed data show that both grow with M , but at different rates and with different signs in the small- M regime.

For comparison, the non- D covariance share $S(M) = (A + X + Y) / \text{Cov} = 1 - D / \text{Cov}$ increases from approximately 1.5% at $M = 6$ to 60.4% at $M = 9,699,690$ (Table 3). This should be read as an algebraic decomposition share, not as a causal fraction of dependence removed by conditioning. The 3.74% between-phase variance share and the 60.4% non- D covariance share describe different one-point and two-point statistics and are not directly comparable effect sizes.

6.4 What this section does not claim

The variance split is an algebraic identity. It partitions the detrended variance into a between-phase and a within-phase term; it does not attribute a causal mechanism to either. The phase-conditional mean $\mu(r)$ is a computed quantity, not a modelled one.

No claim is made that the residual lag-one correlation ρ_{resid} converges to zero, nor that the between-phase variance converges to a finite limit, nor that the phase decomposition explains all of the observed dependence. The empirical claims are those stated in Sections 4 and 5: the decomposition is exact, the residual remains negative at all seven tested moduli, and the pre-registered ranges are reproduced at 10^{10} .

7 Theory

7.1 Scope of the theoretical discussion

The empirical results of Sections 3–6 describe a measured phase decomposition of the lag-one covariance of locally detrended prime gaps. They do not by themselves identify a mechanism. This section states what a Hardy–Littlewood description of the phenomenon would look like, and states clearly what we do not claim.

7.2 Sieve-phase dependence

The observed dependence of the detrended gap on the opening-prime residue modulo a primorial is consistent with the general arithmetic principle that local congruence constraints modify the density of prime tuples. In a Hardy–Littlewood description, these local effects are encoded by singular-series factors. For consecutive gaps $g = h$, one must additionally exclude configurations containing intermediate primes; this is precisely the role of an inclusion–exclusion correction on top of the pair-density heuristic.

A finite inclusion–exclusion calculation of this kind can provide a heuristic estimate of the consecutive-gap distribution. We do **not** use such a calculation to define our estimators or to claim an asymptotic prediction. It is mentioned here only to record that a natural theoretical counterpart of the empirical phase decomposition exists, and that it has not been used in the present paper.

The cross term $Y = \mathbb{E}[u_n \mu(r_{n+1})]$ is the component most directly connected to the consecutive-residue viewpoint of Lemke Oliver and Soundararajan, because the next phase class satisfies $r_{n+1} \equiv r_n + g_n \pmod{M}$. This is a conceptual connection, not an identification of Y with their transition-bias statistic.

7.3 The measured offset d

The quantity

$$d = L - \text{sd}(e)$$

is reported as a **finite-range measurement** over the window $[10^{10}, 3 \times 10^{10})$. Its observed value $d = 2.631$ lies in the pre-registered band $[2.5, 2.7]$ for this window (Section 5), consistent with the value observed at 10^9 in Berramdane (2026a).

The observed level of d , and its drift across logarithmic windows, should not be identified with a theorem-level constant. The present data establish the **reproducibility** of the phase-conditioned covariance decomposition; the derivation of the absolute value and the drift of d from first principles remains an open question.

7.4 Relation to existing heuristics

The relation to existing prime-in-interval variance heuristics is **qualitative**. Those frameworks concern cumulative counts or interval variances, whereas the present observable is the lag-one covariance of consecutive numerical gaps after local detrending. Any transfer of results between the two settings would require additional arguments, which are not supplied here.

Similarly, the Montgomery–Soundararajan framework provides a conditional variance-consistency benchmark for a cumulative correlation statistic. Applying that framework to a fixed number of consecutive gaps is an extrapolation beyond its stated asymptotic regime and is not made in this paper.

7.5 What is not claimed

We do not claim:

- that the phase decomposition explains all of the observed dependence;
- that the residual correlation ρ_{resid} converges to zero as M grows;
- that a Hardy–Littlewood or Montgomery–Soundararajan calculation proves the observed value of d ;
- that the present work constitutes a new theory of prime gaps;
- that the results extend beyond the tested finite range.

We claim only that the phase-conditioned covariance decomposition is exact and reproducible at 10^{10} , that the residual lag-one correlation remains negative at all seven tested moduli, and that the measured offset d is stable across the pre-registered ranges.

7.6 A finite inclusion–exclusion calculation as future work

A finite truncated inclusion–exclusion calculation for intermediate-prime exclusion is a possible diagnostic for the magnitude of d . It is not required for the empirical claims of this paper and is not presented here as a prediction or a test. If it is performed in future work, it should be pre-registered with explicit choices of pair-density normalization, intermediate-offset truncation depth, tail treatment, and renormalization conventions, and reported as a **finite truncated inclusion–exclusion diagnostic** rather than as a hypothesis test.

Related work: Murray (SSRN 7426882) develops inclusion–exclusion formulas with pair-closed truncation and frozen measurements. The present paper does not adopt those conventions and does not compare numerical values; the reference is recorded for context.

8 Limitations

The present study is a finite-range empirical analysis. The results reported in Sections 3–6 are conditional on the specific window, the specific detrending, and the specific ladder of moduli chosen. The limitations below are stated so that no reader mistakes the reported decompositions for asymptotic statements.

8.1 Finite window

The analysis is confined to the single window $[10^{10}, 3 \times 10^{10})$, containing 844,953,415 consecutive prime gaps. No claim is made about the behaviour of the lag-one covariance outside this range. The agreement with the pre-registered predictions (Section 5) is an out-of-sample consistency check at a fixed scale, not evidence of an asymptotic law.

8.2 Finite ladder of moduli

Seven primordial moduli are used:

$$M \in \{6, 30, 210, 2310, 30030, 510510, 9699690\}.$$

The monotone growth of `var_between` and the monotone decline in magnitude of ρ_{resid} across this ladder are observed facts about these seven points. No extrapolation to $M \rightarrow \infty$ is supported. In particular, the data do not establish that $\rho_{\text{resid}}(M) \rightarrow 0$, nor that it converges to any nonzero limit.

A note on finite-sample estimation. At $M = 9,699,690$ the number of phase classes is $\varphi(M) = 1,658,880$, with an average of 509 gaps per class and a smallest class of 433. The phase classes form an exact partition, so the decomposition itself is not affected by empty classes. Finite class sizes nevertheless affect the stability of the estimated conditional means and hence of D , especially at the largest modulus. We therefore interpret the largest-modulus result as a high-resolution diagnostic rather than as an independent validation. The observed decline in $\rho_{\text{resid}}(M)$ is not asserted to be free of finite-class estimation error.

8.3 No causal claim

The decomposition

$$\text{Cov}(e_n, e_{n+1}) = A + X + Y + D$$

is an algebraic identity. All four terms are computed from the same data. The decomposition partitions the covariance; it does not identify a causal mechanism. In particular, the phase component A should not be read as “the cause” of the observed covariance, and the residual term D should not be read as “the true dependence” stripped of phase effects.

8.4 Detrending is a modelling choice

The local detrending $g_n \mapsto e_n = g_n - (\alpha + \beta \log p_n)$ is a specific modelling choice, frozen in the protocol and inherited from Berramdane (2026a). A different detrending would produce different residuals and hence a different decomposition. The consistency of the present results with the frozen 10^9 predictions is a check on this choice, not a proof that it is unique.

8.5 Phase conditioning is on the opening prime only

Conditioning is performed on $r_n = p_n \bmod M$, the residue of the opening prime. The closing prime p_{n+1} also has a residue modulo M , and joint conditioning on both residues would produce a finer decomposition. The present analysis does not pursue this refinement; the observable of interest is the lag-one covariance of numerical gap sizes, conditioned on the opening residue.

8.6 Bootstrap scope

The 95% intervals for ρ_{resid} are obtained by a moving-block bootstrap with 40 contiguous blocks and 1000 replicates. The block bootstrap is appropriate under the assumption that the residual sequence is approximately stationary within the window and that the block length is large compared to the dependence range. The intervals are conditional on these choices. They are not i.i.d. bootstrap intervals, and they do not constitute a test of stationarity.

8.7 No relation to HL/MS as a theorem

The present results are compatible with a sieve-phase interpretation of the lag-one covariance, but they do not prove the Hardy–Littlewood conjecture, the Montgomery–Soundararajan law, or any asymptotic statement about prime gaps. The Montgomery–Soundararajan framework provides a

conditional variance-consistency benchmark for a cumulative correlation statistic; applying it to a fixed number of consecutive gaps is an extrapolation beyond its stated asymptotic regime, and is not made here.

8.8 No comment on Murray

This paper does not evaluate, extend, or contradict the results of Murray (SSRN 6947578, SSRN 7426882). The distinction between the two observables is recorded in Section 9. No comparison of numerical values is made, because the observables are different.

8.9 The IE diagnostic is optional future work

The inclusion–exclusion (IE) diagnostic discussed in Section 7.6 is a possible future-work heuristic. It is not used to define the observable, and its outcome would not modify the definitions of A , X , Y , or D . A positive or negative IE result would change the theoretical interpretation, not the measured decomposition.

8.10 Reproducibility

All numerical values in this paper are computed from the frozen 10^{10} run. The full source, the full output log, the CSV, and the SHA256 checksums are archived on GitHub (commit `bb3548c`, `paper2/SHA256SUMS`). No value is reported without a corresponding run. The small-scale validation at 10^6 and 10^7 passes before the large run; the two-pass streaming implementation uses 64-bit prime coordinates and does not reconstruct primes from floating-point logarithms.

9 Relation to Murray

Murray studies recurrence statistics of consecutive-prime residue classes under CRT-aligned primorial lifts, using finite-state transition models and projections back to a coarse modulus. Our observable is different: we study the lag-one covariance of numerical prime-gap sizes after local detrending, conditioning on the opening-prime residue modulo a primorial. The two approaches are complementary: both resolve hidden divisibility coordinates, but they address different observables and make different statistical claims.

No numerical comparison with Murray’s results is made in this paper. The purpose of this section is to record the difference of observable, not to evaluate or extend Murray’s claims.

A note on terminology. Murray’s second paper uses the term “covariance” for the resolution covariance of tuple densities under primorial lifts. In the present paper, “covariance” refers throughout to the statistical lag-one covariance of consecutive gap sizes after local detrending. The two usages are unrelated.

References

- [1] R. Berramdane. *Persistent Negative Lag-One Dependence in Locally Detrended Prime-Gap Windows up to $N = 10^9$* . 2026a. DOI 10.5281/zenodo.23162269.
- [2] H. Cramér. On the order of magnitude of the difference between consecutive prime numbers. *Acta Arithmetica*, 2:23–46, 1936.
- [3] P. X. Gallagher. On the distribution of primes in short intervals. *Mathematika*, 23(1):4–9, 1976.
- [4] A. Granville. Harald Cramér and the distribution of prime numbers. *Scandinavian Actuarial Journal*, 1995(1):12–28, 1995.

- [5] F. B. Holt. *Expected biases in the distribution of consecutive primes*. arXiv:2405.03540, 2024.
- [6] R. J. Lemke Oliver and K. Soundararajan. *Unexpected biases in the distribution of consecutive primes*. *PNAS*, 113(31):E4446–E4454, 2016. arXiv:1603.03720.
- [7] H. L. Montgomery and K. Soundararajan. *Primes in short intervals*. *Communications in Mathematical Physics*, 252(1-3):589–617, 2004. arXiv:math/0409258.
- [8] A. Ash, L. Beltis, R. Gross, and W. Sinnott. Frequencies of successive pairs of prime residues. *Experimental Mathematics*, 20(4):400–411, 2011.
- [9] W. Banks, K. Ford, and T. Tao. *Large prime gaps and probabilistic models*. *Inventiones Mathematicae*, 233:1471–1518, 2023. arXiv:1908.08613.
- [10] D. J. Murray. *Prime-Residue Projection Tomography of Consecutive-Prime Biases: Primorial Recovery and Gap-Word Order Asymmetry*. SSRN 6947578, 2026.
- [11] D. J. Murray. *Recursive Predictive Closure in the Prime Sieve: Resolution covariance, finite probability consistency, and gap-word prediction*. SSRN 7426882, 2026.

10 AI Disclosure

This work benefited from AI assistance. The author used Claude (Anthropic) for theoretical guidance, editorial review, and manuscript preparation. Kimi was used for critical review and references. A local execution assistant was used for file management and LaTeX compilation. All mathematical content, data analysis decisions, and final claims are the author’s sole responsibility. No AI tool was used to generate or alter the numerical results reported in this paper; all numerical results were computed by the archived code at `source/data` commit `bb3548c`.